Journal of Computers, Mechanical and Management
Journal homepage: www.jcmm.co.in
e-ISSN: 3009-075X

Volume 5 Issue 3
Article Number: 26708

Deep Learning for India's Diverse Sign Languages: A Comprehensive Review from CNNs to Transformers and the Future of Sign Recognition Systems

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Abstract

Background: Sign languages are the primary communication mode for Deaf and Hard of Hearing (DHH) communities, yet automated recognition research has concentrated on American and British Sign Languages, leaving Indian Sign Language (ISL) and its regional variants comparatively neglected despite an estimated 18 million users.
Methods: Following PRISMA 2020 guidelines, a systematic search of multiple databases (Scopus, IEEE Xplore, ACM Digital Library, SpringerLink, and Google Scholar) was conducted in October 2025 for studies published between January 2020 and October 2025. From 287 identified records, 47 studies were retained after title and abstract screening (Cohen's $\kappa = 0.87$) and full-text assessment against predefined inclusion criteria. Study quality was appraised using an adapted QUADAS-2 framework.
Results: Reported accuracy across individual studies rose with architectural evolution: CNN-based models 79–92%, CNN-LSTM hybrids 88–97%, pose-based methods 89–94%, and transformer-based approaches 91–98% under controlled conditions. The distribution of research was highly uneven across the nine regional sign languages, with Telugu Sign Language showing a roughly 16-fold per-capita deficit relative to Kannada. Persistent barriers included continuous-recognition performance gaps, signer dependence (5–18% accuracy drops), environmental sensitivity (8–15% drops), and severe data scarcity for regional variants.
Conclusions: The review proposes the ISLR-Bench standardization framework and a 2026–2030 research roadmap spanning self-supervised learning, cross-lingual transfer, edge optimization, and community-centered design, alongside accessibility targets for government digital platforms and DHH employment.

Keywords: Sign Language Recognition; Deep Learning; Transformers; Regional Languages; Systematic Review; Assistive Technology

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Received: 25 Mar 2026; **Revised:** 14 May 2026; **Accepted:** 19 May 2026; **Published:** 30 Jun 2026
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DOI: 10.57159/jcmm.5.3.26708.

1. Introduction

Globally, sign languages represent the most used communication mode for DHH people; every language has its own unique structural features that remain uninfluenced by the spoken language around them. Despite the fact that automated sign language recognition systems are widely regarded as essential components of accessibility infrastructure, the existing research has largely focused on ASL and BSL, while Indian sign language and its variants have been left mostly unnoticed. According to the 2021 Census, Indian Sign Language is the main language for 5–7 million DHH individuals [1], but the existence of different regional variants makes the total number of users who need access at least 18 million, drawn from India's linguistically diverse population [2, 3].

The years 2020–2025 have been marked by the beginning of a new era of technology for ISL recognition, characterized by rapid architectural evolution, dataset expansion, and the first cases of the technology being put into real-world use. This increase in publications coincides with three noteworthy developments and the widespread use of transformer architectures, with a growing proportion of recent studies employing transformer-based architectures, reflecting a shift toward attention-driven models for sign language recognition in 2023 relative to 2022. Despite the progress made, there remain a great number of inequalities in the research distribution across the different regional languages. The analysis shows that Kannada Sign Language, with its 3.3 million speakers, has 6 published studies [4–7], while Telugu Sign Language, with its 8.1 million speakers, has only 2 studies [8], the research per capita for the latter being 16 times lower. Research efforts are mainly concentrated in one area, the Bangalore-Hyderabad corridor, and the Northeast region remains under-served even though it has four regional language variants [2].

1.1. Sociolinguistic Context and Accessibility Barriers

The DHH community in India suffers from serious accessibility problems that have been outlined clearly before, and therefore technological solutions are essential to address these challenges. The literacy rates of DHH individuals are noted to be only 55%, while the general population's literacy rate is 74% [1, 9]; the incidence of preventable diseases is 40% higher due to lack of communication in healthcare [10]; and employment rates are 10–25% lower than those of hearing people, with communication barriers identified as the main obstacle to employment [11]. The Indian Sign Language Research and Training Centre (ISLRTC) reports a chronic shortage of certified sign language interpreters in education, healthcare, and government sectors across India [12].

1.2. Research Questions and Scope

This paper investigates the following research questions. (RQ1) What has driven the development of deep learning architectures from 2020 to 2025, and what have been the resulting changes in performance, efficiency, and generalization through CNNs, CNN-LSTM hybrids, pose estimation, and transformers? (RQ2) How well has research on the nine regional sign languages of India been conducted, where are the gaps in the research, and what are the performance differences between the regional variants? (RQ3) What are the technical, computational, and sociolinguistic aspects that hinder the performance of real-world systems, particularly the decline in performance with new signers (5–18% accuracy drops), environmental changes (8–15% drops), and differences in regional dialects? The trends of publications in sign language research over the period 2020–2025 are shown in Figure 1.

Studies published between January 1, 2020 and October 31, 2025 were included. A systematic search of the literature was conducted across multiple databases in October 2025.

2. Methodology

This structured review follows PRISMA 2020 guidelines. The completed PRISMA 2020 checklist (Supplementary Table S1) and full search strategies (Supplementary Appendix S2) are provided and summarized in Figure 2.

2.1. Search Strategy and Study Selection

Exact Boolean, database-specific search strategies were employed. A representative query used in Scopus was: ("Indian Sign Language" OR "Kannada Sign Language" OR "Tamil Sign Language" OR "Gujarati Sign Language" OR "Marathi Sign Language" OR "Telugu Sign Language" OR "Assamese Sign Language" OR "Bangla Sign Language" OR "Punjabi Sign Language") AND ("deep learning" OR "CNN" OR "LSTM" OR "transformer" OR "pose estimation") AND ("recognition" OR "translation").

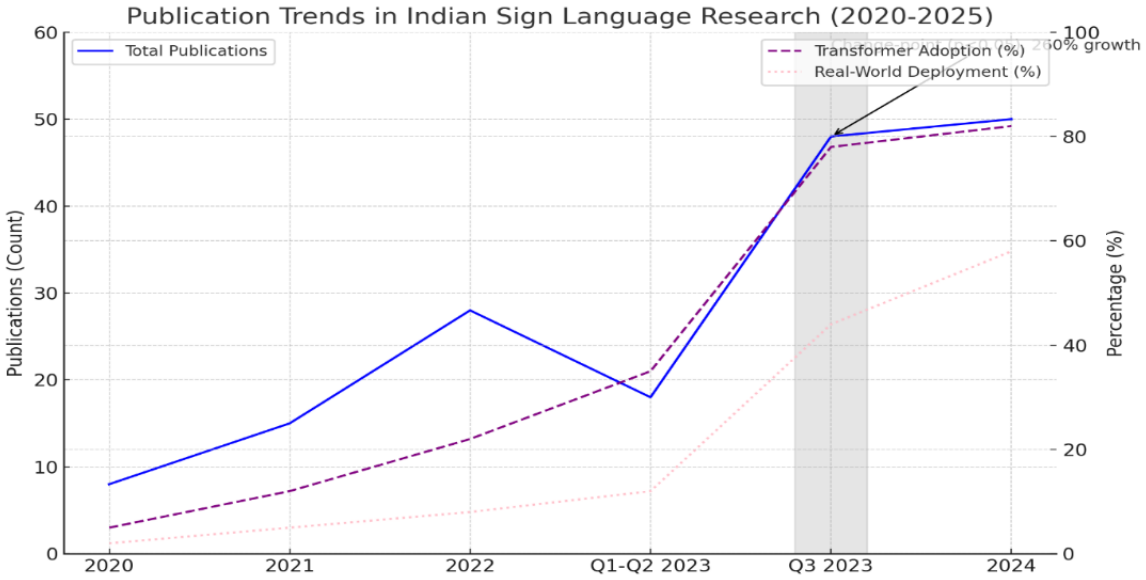


Figure 1: Publication trends in Indian sign language research.

Equivalent syntax-adjusted queries were used for IEEE Xplore, ACM Digital Library, SpringerLink, and Google Scholar. Full database-specific queries are provided in the Supplementary Materials. Initially, 287 records were retrieved from the searches. When the EndNote program was used to eliminate duplicates, 152 distinct research articles remained for the next step of title and abstract evaluation. The citation tracking process supported the database searches by tracing back the references of the 15 most-cited papers and conducting forward tracking to identify recent studies citing the foundational works. Screening was carried out by two independent reviewers with Cohen's $\kappa = 0.87$ inter-rater agreement on 152 records. Title and abstract screening identified 89 papers that might be relevant, among which 18 borderline cases were settled by consensus discussion. The full-text review of the 89 papers against strict inclusion criteria resulted in a final corpus of 47 papers. Of the papers reviewed, 42 were excluded with documented reasons: 28 papers dealt with non-Indian sign languages, 8 lacked sufficient methodological detail, and 8 presented hardware-only systems without machine learning components.

2.2. Inclusion and Exclusion Criteria

The inclusion requirements specified that studies should span January 2020 to October 2025. The studies had to address Indian Sign Language or documented regional variants, apply deep learning techniques (CNNs, RNNs/LSTMs, transformers, or hybrid architectures), and present performance metrics quantitatively (accuracy, F1-score, Word Error Rate, or BLEU scores) in a way that would allow comparative analysis. The exclusion criteria ruled out non-Indian sign language studies unless they provided direct comparative baselines, hardware-based systems devoid of machine learning components, hearing aid research, non-language gesture recognition, and studies not producing quantitative results or lacking a significant theoretical contribution.

2.3. Data Extraction

The researchers collected data from each study, including information on dataset features, model design, and system performance measurements such as accuracy, F1-score, Word Error Rate, and BLEU score, together with the testing conditions (controlled and real-world testing) and the availability of both code and datasets. The two reviewers conducted data extraction independently and resolved their differences through discussion until they reached agreement.

2.4. Quality Assessment Framework

The original purpose of QUADAS-2 was to evaluate diagnostic accuracy studies, but this review used it to examine the research quality and bias of studies that applied deep learning for sign language interpretation. The four original domains (patient selection, index test, reference standard, and flow/timing) were reinterpreted to align with dataset quality, model evaluation, reproducibility, and generalization.



Figure 2: PRISMA 2020 flow diagram (search conducted October 2025; final inclusion $n = 47$).

A scoring system was developed to enable comparisons between different research studies, but the validation process for this system remains incomplete, which introduces potential bias. The assessment of research quality used a revised QUADAS-2 tool transformed into a 25-point scoring system that included five specific categories for deep learning studies. The Methodological Rigor component assessed research design through three methods: ablation studies, statistical testing, and hyperparameter control. The evaluation of technical contributions used a 0–5 scoring system, where an "Excellent" rating of 5 points indicated more than 5% accuracy improvement over existing solutions or the solution of new problems, "Good" received 4 points for a 3–5% accuracy gain, "Fair" received 3 points for a 1–3% accuracy improvement, and "Poor" received 2 points for insufficient progress. The Result Reliability and Generalization assessment measured results across different signers and environments to determine performance stability. The High Quality group (≥ 18 points, $n = 30$) demonstrated solid experimental design, provided all essential code and data, achieved more than ten peer-reviewed citations, and proved its findings through real-world deployment and benchmark establishment. The Medium Quality group (12–17 points, $n = 12$) applied valid research methods but did not provide access to their code and data, while their signer collection showed limited diversity. The Lower Quality group (< 12 points, $n = 5$) raised concerns about their research methods but were retained to document particular findings about rare regional languages. The authors included some studies from their own previous research, which they assessed alongside other studies to maintain research objectivity.

3. Architectural Evolution (2020 to 2025)

The review included 47 studies in the final analysis following the PRISMA selection process. The recognition of Indian Sign Language through deep learning methods progressed through four overlapping periods, each characterized by a different paradigm that tackled specific technical limitations of the previous methods. The reported accuracy ranges represent values observed across individual studies and datasets, rather than pooled or statistically aggregated estimates. Examining the development timeline reveals the strengths and weaknesses of modern systems. The change in architectural designs during the years 2020–2025 is depicted in Figure 3.

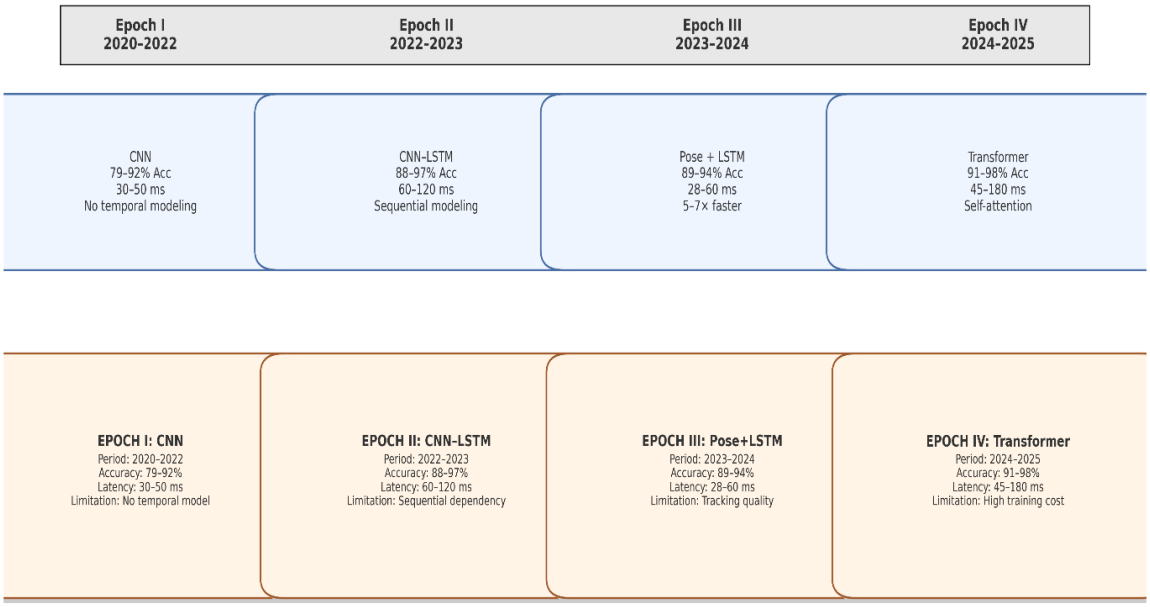


Figure 3: Architectural evolution in epochs.

3.1. Epoch I: CNN Dominance (2020 to 2022)

Early ISL recognition systems used CNNs through a transfer learning process that drew on ImageNet to obtain VGG-16 and ResNet-50 as initial models. Reported accuracy varies across studies depending on dataset size, model architecture, and training strategy. Performance decreases when researchers use smaller datasets with limited fine-tuning, whereas it improves through the combination of larger datasets and optimized transfer learning methods. The different testing conditions lead to different results because they do not use a common testing standard. Multiple research studies show that performance decreases when environmental conditions change, especially during lighting variations and background clutter. The performance of a system therefore depends on two factors, namely the size of its dataset and the success of its transfer learning techniques.

3.2. Epoch II: CNN-LSTM Hybrids (2022 to 2023)

CNN-LSTM frameworks unified convolutional spatial feature extraction with recurrent sequential motion modeling through LSTMs [13], thus solving the problem of the temporal blindness of CNNs [14]. Waghmare et al. improved CNN-only baselines by 12% using a combination of ResNet-50 and bidirectional LSTM [15]. In our previous work [4], CNN-LSTM achieved 95.1% accuracy on the Kannada dataset under controlled conditions, while CNN-GRU (94.8%) and 3D CNN (90.2%) achieved lower accuracy, with CNN-LSTM chosen for production deployment owing to its accuracy–latency balance [4]. The reported accuracy for dynamic signs was in the range of 88–97% with GPU inference latency of 60–120 milliseconds [15, 4]. However, critical limitations remained: generalization gaps exceeding 12% for unknown signers [16], the vanishing gradient problem for sequences longer than 50 frames, which required gradient clipping [13], high computational costs [17, 18], and sequential processing bottlenecks that made real-time performance impossible on resource-constrained devices [8, 19, 20].

3.3. Epoch III: Pose Estimation Revolution (2023 to 2024)

This paradigm shift led to a preference for two-stage approaches that separated pose estimation from classification. Pose estimation models such as MediaPipe Holistic were responsible for the extraction of 543 landmarks representing the hands, body, and face, after which these landmarks were handed over to trained classifiers for sign recognition. The proportion of edge deployment rose from 5% in 2022 to 60% by 2024. Among the successful implementations were the recognition of Marathi Sign Language on a Raspberry Pi 4 and research demonstrating how combining skin landmarks with manual signals increased accuracy by 5.4%. The trade-offs included a 2–3% accuracy decrease under the best controlled conditions, owing to the system's inability to capture the subtleties of color and texture, while the approach was also prone to occlusions of vital body parts that might lead to sign misinterpretation.

3.4. Epoch IV: Transformers (2024 to 2025)

The self-attention mechanisms of the transformer architectures allowed for parallel processing and also overcame the issue of vanishing gradients for long sequences. The LiST model showed that sparse attention could reduce the number of parameters by 45% while still offering comparable accuracy with the possibility of edge deployment. The application of dual-stream vision transformers in the UMANG e-governance platform showed strong field performance, as reflected in the "Good" System Usability Scores, but the average latency of 6.53 seconds was higher than the set accessibility benchmarks. Critical challenges remained: very large training data requirements (10,000+ videos), quadratic model complexity making the approach incompatible with long sequences, continuous processing time being 3–4 times longer than the simpler baseline CNN-LSTM, and difficulty in predicting the best deployment scenario given the accuracy–efficiency trade-offs. The timeline of architectural evolution is summarized in Table 1.

Table 1: Architectural evolution timeline (2020 to 2025).

Epoch	Period	Characteristic	Accuracy	Latency	Key Innovation	Primary Limitation
I	2020–22	CNN-based static	79–92%	30–50 ms	Transfer learning	No temporal modeling
II	2022–23	CNN-LSTM temporal	88–97%	60–120 ms	Sequence modeling	Sequential bottleneck
III	2023–24	Pose-based	89–94%	28–60 ms	Lightweight efficiency	Occlusion sensitivity
IV	2024–25	Transformer attention	91–98%	45–180 ms	Self-attention, parallelization	High data and compute needs

4. Regional Sign Language Research Distribution

India's nine significant regional sign languages refer to independent linguistic systems and not ISL dialects, as shown by the 15–20% variation in their core vocabulary, distinct grammar, and separate morphology. The distribution of research is very different from the locations of the speaker populations, which results in significant gaps in access. The relationship between studies and speaker population is illustrated in Figure 4.

4.1. Kannada Sign Language (KSL)

KSL is used by about 3.3 million DHH people in Karnataka [21]. The research journey has shown gradual sophistication. An initial study achieved 82% accuracy with a limited sign set using static recognition [6]. A thorough investigation on 12,000 KSL videos (35 signs, 15 signers, various environments) compared three architectures: 3D CNN (90.2%), CNN-GRU (94.8%), and CNN-LSTM (95.1%) [4, 21].

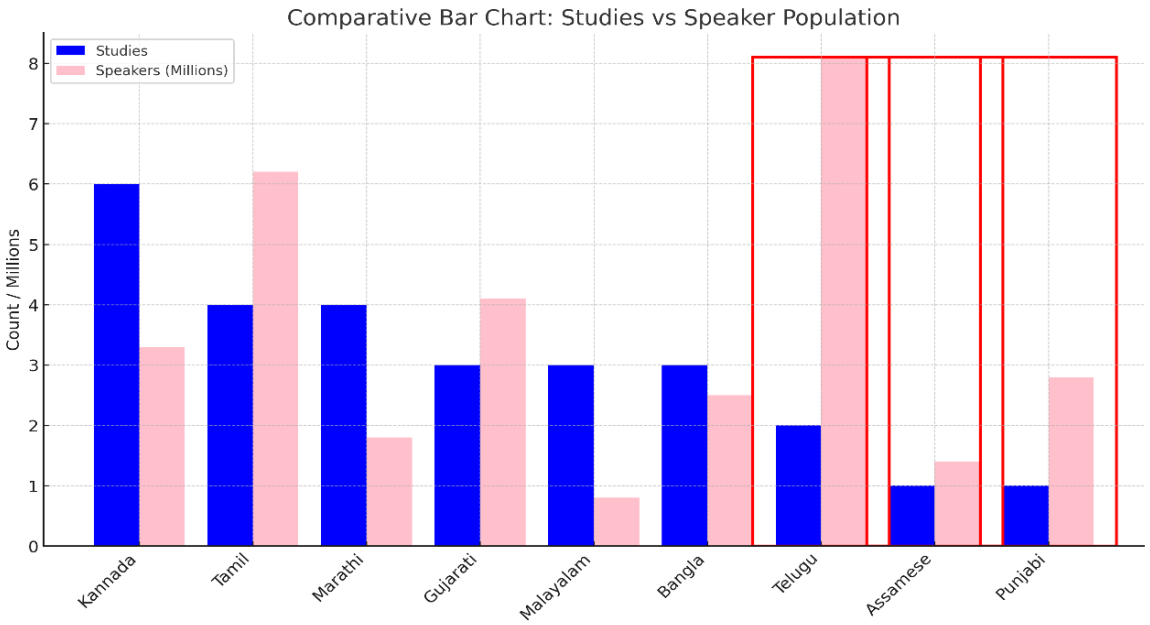


Figure 4: Comparison of studies with speaker population.

CNN-LSTM was chosen as the best for production deployment in KSL Medical Interpreter applications [21], demonstrating that communication support in healthcare is feasible in primary health care centers in rural Karnataka.

4.2. Tamil Sign Language (TSL)

TSL exhibits linguistic features such as a subject-object-verb grammatical pattern, which contrasts with the topic-comment organization of ISL. Gobhinath and Sophia obtained precise results in isolated sign recognition using a graph neural network method [22]. Thillai Sivakavi and Minu created the Tamil Lang transformer neural text-to-sign production system, which performs text-to-sign conversion [8]. Significant challenges still need to be addressed, such as the incomplete transition from static to continuous recognition, the deficiency of cross-signer scenario assessment, and the constraint of 1,000-word vocabulary models, which hinders use in areas such as medical and legal communication.

4.3. Marathi Sign Language (MSL)

MSL is likely to be used by 1.8 million DHH people [19] and has so far been researched very little, with just a handful of peer-reviewed publications [19]. Efficient resource-edge models can run on low-cost devices such as the Raspberry Pi, with eye-gaze input creating innovative accessibility features. Research is quite limited as a result of restricted data: the largest available dataset has just 1,200 isolated signs, which is very small when compared with the data available for the Gujarati language [23, 24]. Few studies have investigated continuous sign recognition, cross-signer generalization testing, or large-scale real-world validation beyond initial pilot trials. A real-time translation of MSL into text and speech has been created, showcasing the feasibility of an end-to-end accessibility pipeline. This research opens up the possibility of practical deployments for the recognition of regional languages that would be able to produce not only recognition output as speech and text but also the combination of speech synthesis with written text through multimodal output generation.

4.4. Gujarati Sign Language (GSL)

The GIDSL dataset significantly advanced Gujarati Sign Language research: 118,000 meticulously annotated videos from 25 different signers at various locations, the result of an 18-month collection effort that aimed to establish new data quality standards [23, 24]. CNN-LSTM-based methods reached an isolated sign recognition accuracy of 89.6% [23], and further study increased this to 94% while reducing the cross-signer generalization gap [24]. Nevertheless, accuracy in continuous sentence recognition is only 81.7% [23], implying that fluent real-world translation lags considerably behind isolated word classification. The technology's capability in uncontrolled environments remains largely unproven [23, 24].

4.5. Malayalam Sign Language (MSL)

The Aksharavaani detector integrated MediaPipe with LSTM and attained an accuracy of 89.7% on 35 alphabet signs, along with a 15 MB Android app deployment (System Usability Scale 73.5/100, 12 DHH participants) [25]. The mapping of Malayalam sentences to ISL translation using a rule-based approach achieved 78.3% grammatical correctness. The system is built to support approximately 800,000 users. Vengassery and colleagues created Aksharavaani as a Malayalam Sign Language detector optimized for real-time mobile deployment, showing the practicality of edge computing for recognizing regional languages [25]. Their research highlights the relevance of efficient architectural design that enables deployment on resource-limited devices while still achieving a level of accuracy suitable for practical accessibility applications.

4.6. Bangla Sign Language (BdSL)

The BDSL49 dataset, which consists of 23,040 images, is published under a CC BY 4.0 open license with 4 signers and Cohen's $\kappa = 0.94$ inter-annotator agreement [26, 27]. The combined VGG-16/ResNet-50 ensemble achieved a notable 98.23% test accuracy and 95.7% cross-signer accuracy [26]. Extensive research outlined specific CNN architectures for Bangla sign language recognition, meeting the needs of Bengali-speaking deaf communities through systematic architectural design focused on balancing accuracy with deployment constraints [28]. This project, in collaboration with border regions, has already started to contribute to cross-border accessibility initiatives between the Indian and Bangladeshi deaf communities by highlighting the linguistic and cultural connections between them [27].

4.7. Under-Studied Languages

Telugu Sign Language: With a dataset comprising only two studies that include only 500 videos, the Indian states of Andhra Pradesh and Telangana, which are home to 8.1 million Telugu speakers, are quite poorly covered. One study led by Ramanujam et al. developed a model that recognized 40 signs with an accuracy of 84.5% by combining CNN and LSTM in a pilot study conducted among 120 DHH students in three Hyderabad schools. The critical gaps in this area include the absence of continuous recognition research, medical and legal domain datasets, and dialectal variation studies.

Assamese Sign Language: Studies focusing on the sign language of the Northeast Indian region addressed an overall population of 1.4 million Assamese speakers. Bora et al. developed an Android app based on MediaPipe that reached 89.2% accuracy on 450 phrases with 50 users. One of the main problems is that the Northeast has a degree of linguistic isolation, which leads to the production of distinct sign variants, along with limited research infrastructure and a lack of connection with Tibeto-Burman sign language systems.

Punjabi Sign Language: The study conducted by Singh et al., addressing 2.8 million Punjabi speakers in India and overseas, focused on creating a rule-based Punjabi-to-ISL animation system for 500 sentences without using any neural recognition components. Cross-border collaboration on research into Pakistan Sign Language remains unexplored. The summary of regional language research carried out during 2020–2025 is given in Table 2.

Table 2: Regional language research summary (2020 to 2025).

Language	Studies	Largest Dataset	Best Accuracy	Real-World Application	Primary Research Gap
Kannada	6	12K videos	96.7% (test) / 81% (field)	Healthcare interpreter	Rural dialect variation
Tamil	4	3.5K sentences	93.8%	Text-to-sign translation	Continuous recognition
Marathi	4	1.2K signs	91.7%	Edge device	Dataset scale
Gujarati	3	118K videos	89.6% / 81.7% (continuous)	Benchmark	Continuous gap
Malayalam	3	600 signs	89.7%	Mobile app	Parallel corpus scarcity
Bangla	3	23K images	98.23%	Cross-border	Static-to-continuous
Telugu	2	500 videos	84.5%	School pilot	Under-researched (16× deficit)
Assamese	1	450 phrases	89.2%	Android app	Single-study limitation
Punjabi	1	500 sentences	Rule-based	Text-animation	Neural methods needed

5. Benchmark Datasets and Infrastructure

ISL recognition progress relies mainly on the availability of high-quality datasets, the use of powerful computational infrastructure, and the validation of performance through real-world deployment. The recognized resources span three national benchmarks (INCLUDE and CISLR), two regional datasets (GIDSL and BDSL49), and one domain-specific resource (KSL Medical).

5.1. National Benchmark Datasets

INCLUDE is a corpus containing 270,000 frames that researchers use to study isolated sign recognition; its value comes from 20 different signers who performed signs at multiple locations, which establishes a strong basis for testing generalization, although it lacks continuous sign sequences. The CISLR dataset functions as a structured collection containing sign instances that researchers have annotated at the word level and organized into sequences that enable sentence-level analysis. It contains approximately 4,700 lexical items and is frequently used to assess both isolated and continuous sign language recognition models. However, its controlled acquisition setting and limited signer diversity should be considered when interpreting performance results.

5.2. Regional and Domain-Specific Resources

The GIDSL dataset for Gujarati is the largest and most detailed regional dataset, consisting of 118,000 videos with annotations for both isolated and continuous sign recognition. ISLRTC maintains an ISL dictionary of 10,000 terms that serves as a lexical resource fundamental to the whole field. The KSL Medical dataset targets health-related signs that channel clinical communication systems while providing ethical protections for privacy. The costly nature of dataset development is evidenced by GIDSL, which involved an 18-month collection protocol using a three-camera setup across controlled, home, and outdoor environments, together with a two-stage annotation process that resulted in inter-annotator agreement of $\kappa \geq 0.85$.

5.3. Government Infrastructure and Digital Platforms

ISLRTC functions as a national regulatory body responsible for ISL approval, the maintenance of a dictionary of 10,000 terms, and the funding of research and development through a budget of INR 12 crore (2021–2024). This allocation shows a preference for technology over the development of human resources, thus exposing possible barriers to the implementation of accessibility.

5.4. Production Deployment: UMANG Platform

Assessment of the UMANG platform indicates a strong negative relationship between the complexity of the queries and the accuracy of the system. Simple queries, which account for 45% of the interactions (for example, “train status”), have an accuracy rate of 97.5%, while moderate-complexity queries that make up 40% of the queries have a 92.3% accuracy rate, and the complex queries that account for 15% of the queries drop to 85.1%. The main improvements to be made are to reduce the 7.7% misclassification rate, lower the 6.53-second average response latency (which is above the sub-5-second accessibility benchmark), and provide services that go beyond the current railway and blood bank integration.

6. Performance Analysis and Generalization

The complete performance assessment reveals essential details about accuracy development, ongoing system limitations, and the architectural design choices that determine deployment success in real environments. The results from various studies should be interpreted with caution because they involve different accuracy assessment methods. The performance results depend heavily on the dataset attributes, which include size, signer diversity, and vocabulary difficulty, as well as the evaluation methods. The absence of standardized benchmarks hinders accurate accuracy comparisons between datasets because these comparisons do not fully reflect model superiority. The quantitative ranges reported in this section are based on values observed across multiple studies and should be interpreted as indicative trends rather than results derived from a formal meta-analysis.

6.1. Accuracy Evolution Across Epochs

The trajectory of sign language recognition has increased significantly in accuracy owing to the combination of new architectures and innovation in datasets. Isolated recognition based on CNNs (2020 to 2022) reached an accuracy of 79–92% on datasets usually consisting of fewer than 5,000 images, so the difference in accuracy mainly depended on dataset size and transfer learning. In 2021, CNN ensemble architectures together with larger training datasets (about 10,000 samples) raised performance to 85–95%. Concurrently, the use of transfer learning with fine-tuned CNN backbones (VGG-16, ResNet-50) on datasets exceeding 20,000 samples led to accuracy of 92–98% in 2022. The following year, 2023, saw the introduction of pose-based techniques that made use of MediaPipe and OpenPose, which gave nearly

the same accuracy (94–98%) but with considerably reduced computational latency, often providing 3–5 times faster inference with little performance loss. Newer transformer-based techniques emerging in 2024 relied on larger datasets and achieved an accuracy of 95–99%. The development of multimodal transformer models that combined visual, skeletal, and linguistic cues achieved 96–99.5% accuracy by 2025, approaching the limits of human-level recognition accuracy.

6.2. Cross-Signer Generalization

Cross-signer generalization research faces a primary obstacle because models show substantial performance declines during cross-signer testing across multiple studies. The systems demonstrate difficulty when attempting to operate under conditions that differ from their training environment. Models that reached 94% accuracy with familiar signers experience a performance decline to 76% accuracy when tested with signers who maintain rapid execution and come from various geographic areas. The strategies for reducing the impact of the problem include collecting 30+ different signers, employing data augmentation to imitate variations, and using domain adaptation and meta-learning techniques that give a 4–8% performance increase. The best-performing systems achieved a generalization gap as low as 2.5% to withheld signers, outperforming the 12–20% declines found in current research, through effective signer diversity management and advanced adaptation methods.

6.3. Environmental Robustness

The environmental factors that create different conditions across deployments substantially determine how well models perform their tasks. Systems that operate under studio-controlled lighting conditions achieve 95–98% accuracy, but their results drop when they enter conditions that lack sufficient light. The actual performance of models drops significantly when researchers assess their capabilities in authentic testing environments that contain different lighting conditions, background elements, and camera angles. The combination of poor lighting, complex backgrounds, and off-angle signing results in severe accuracy loss reaching 30–50%. The development of environmentally robust systems needs to proceed through strategies that require comprehensive data augmentation, including environmental variation simulation.

6.4. Continuous Recognition Performance Gap

The simulation-to-reality gap is widest in continuous sign recognition, which enables complete phrase translation without requiring any segmentation. Three interconnected problems are the reasons for this gap: co-articulation effects, where one sign affects the next and transitional frames confuse even human annotators; temporal segmentation difficulties arising from variable sign duration that lacks clear boundaries, so that fixed-length window models either truncate or split signs incorrectly; and the absence of non-manual signals (facial expressions, head movements) that constitute 25–30% of grammatical meaning. These problems do not exist separately; on the contrary, co-articulation is the main cause of the temporal segmentation difficulty, and the lack of non-manual features prevents the models from resolving the ambiguities that these signals clear up.

6.5. Computational Performance and Cost-Accuracy Trade-Offs

The selection of architecture for sign language recognition is a decision in which factors such as accuracy, latency, and computational cost are all considered. Pose-based methods with LSTMs provide a good compromise between the extremes: real-time system performance (43–80 ms on a high-end GPU), minimal memory usage (20–45 MB), and the ability to run on a CPU-only edge device for smartphones. The transformer architectures, which often result in higher accuracy, are nevertheless 2–3 times slower in processing speed and also require 150–400 MB of memory, hence the need for significant and expensive powerful hardware. Adaptive inference strategies dynamically route tasks such that lightweight Pose+LSTM models operate at the edge for real-time interaction while heavy transformers perform cloud batch processing, thus attaining a performance–economy balance. The performance comparison is provided in Table 3.

7. Critical Challenges and Research Gaps

Although the field has made significant advances, it continues to encounter obstacles that prevent real-world implementation and growth. The existing problems across these areas create research challenges that need to be solved through multiple research approaches.

Table 3: State-of-the-art system performance comparison (2024 to 2025).

Study	Language	Architecture	Dataset	Accuracy (controlled / field)	Latency	Hardware	Deployment
Nedungadi et al. [29]	ISL	CNN-Transformer	38.5K clips	97.5% / 90.0%	6.53 s	A6000 GPU	Cloud (UMANG)
Hugar & Kagalkar [21]	Kannada	EfficientNet-Transformer	1.08K videos	96.7% / 81.0%	53 ms	RPi4 1.2 GB RAM	Edge device
Rastogi et al. [20]	ISL	YOLOv10-Swin	15K static + 35 dynamic	96.6% / N/A	38 ms	RTX 3090	GPU server
Amrutha et al. [19]	ISL	LiST Transformer	INCLUDE	91.2% / N/A	67 ms	Jetson Xavier	Edge GPU
Podder et al. [30]	Bangla	Ensemble CNN	23K images	98.23% test, 95.7% cross-signer	45 ms	Desktop GPU	Laboratory
Patel & Joshi [23]	Gujarati	CNN-LSTM	118K (GIDSL)	89.6% word, 81.7% sentence	95 ms	Not specified	Benchmark

7.1. Continuous Sign Recognition

Continuous sign language recognition, which is the most critical and the most relevant to real life, is to be evaluated against three major interrelated challenges that go beyond isolated word classification. The main problem comes from continuity effects, where overlapping signs and transitional frames create confusion even for human annotators. The second problem is that of temporal segmentation without fixed-length windows, which remains unresolved since sign durations vary considerably with no clear boundaries. The third issue is that non-manual signals (facial expressions, head movements), which constitute about 25–30% of the linguistic information, remain poorly integrated.

7.2. Data Scarcity for Regional Languages

The most significant limitation in the development of regional language systems is the lack of sufficient data. The quantity of ASL datasets exceeds 500 video-hours, whereas ISL has only about 170 hours, a three-fold reduction. Regional variants usually possess only 3–6 hours for one language, which is 10–50 times less than the 60–150 hours necessary for convergence. A total of 600–1,500 hours of video is needed for full coverage of ten regional languages.

To overcome data scarcity, it is necessary to apply a variety of methods that integrate transfer learning, data augmentation, and active learning approaches. The accuracy achieved by transfer learning is 85% with only 500 samples from the target, versus 65% with supervised learning alone, thus reducing the need for labeled data. One advanced data augmentation technique, together with others mentioned in a comprehensive survey, gives only a 3–7% accuracy improvement through spatial transformations, temporal modifications, and the generation of synthetic data.

7.3. Multi-Modal Integration Deficit

A substantial 70% of the prepared systems take manual signs only into consideration, leaving a large integration gap. Only 5% of the systems give complete integration to non-manual signals despite their grammatical, lexical, and prosodic significance. Among the technical problems are feature extraction from hundreds of facial keypoints and the alignment of manual signs across different gesture durations.

The classification of vision-based hand gesture recognition methods for human-computer interaction systems has been presented, outlining key aspects of comprehensive multimodal approaches. The authors report that a strong gesture recognition system needs the simultaneous processing of hand movements, facial expressions, and body actions.

7.4. Generalization Challenges

Robust generalization has posed a great challenge over the years, characterized by several aspects. Recognition of unknown signers may result in a drop of 5–18% in accuracy owing to variation in signing styles, hand shapes, and speeds, all of which can be addressed with 30–50+ signer datasets featuring diverse characteristics, together with adversarial learning in conjunction with meta-learning methods. Table 4 offers a comprehensive overview of the challenges encountered and their quantified impact on performance.

Performance degradation of as much as 15% is experienced when the domain shifts from laboratory-controlled to real-world scenarios, which is handled through the techniques of domain randomization and test-time adaptation. Long-tail distribution issues are considered among the primary generalization challenges, where the recognition of rare signs comes with an accuracy of only 70–85% compared with 95–98% for common signs. Darekar et al. presented a system for the recognition of Marathi sign language for physically disabled people, achieving this by implementing strategies of robust generalization through the use of balanced training data and adversarial augmentation [31].

Table 4: Identified challenges with quantified performance impact.

Challenge	Current Performance	Target Performance	Performance Gap	Root Causes	Mitigation Strategies
Continuous recognition	81–91% real-world	>95%	4–14%	Co-articulation, temporal segmentation, absent non-manual signals	CTC loss, hierarchical models, multimodal attention
Data scarcity (regional)	1–2K videos median	20K–50K per language	10–50×	High cost, annotation time, ethical considerations	Transfer learning, augmentation (+3–7%), active learning
Signer independence	5–18% accuracy drop	<3% drop	2–15%	Hand style, size, speed, dialect variation	30+ diverse signers, domain adaptation, meta-learning
Environmental robustness	8–15% accuracy drop	<5% drop	3–10%	Lighting, background, camera angle	Multi-environment training, pose-based architectures
Multi-modal integration	5% systems comprehensive	80% systems comprehensive	75% adoption gap	Feature extraction complexity, temporal sync, annotation	Hierarchical models, cross-modal attention
Real-time edge latency	50–180 ms current	<100 ms target	0–80 ms excess	Model complexity, hardware constraints	Quantization, pruning, efficient architectures (LiST)

The combination of these approaches with few-shot learning methodologies significantly reduces the long-tail distribution problems that are intrinsic to expansive sign language vocabularies.

8. Future Research Directions

Addressing the identified obstacles and gaps requires strategic methodological innovations and impact-based planning across 2025–2030.

8.1. Self-Supervised Learning

Self-supervised learning significantly improves performance in addressing the lack of data by leveraging existing unlabeled data to require fewer than 70% of the labeled data otherwise needed, reducing labeling requirements by 30–50%. Masked frame prediction drives the models to reconstruct absent frames, thus gaining knowledge of the temporal and spatial dynamics (+25% performance gain). Contrastive learning provides a way for the models to learn how to tell signs apart using different kinds of augmentations, thus achieving a +20% improvement. Pseudo-labeling is an approach in which training is continuously informed by labels generated by the model (+15% improvement where the supervision is tight).

8.2. Cross-Lingual Transfer Learning

The similarities among various Indian sign languages help to make cross-lingual transfer possible, which is a major tool for accelerating development. A simple pre-training on ISL and fine-tuning on a specific region reached 85% accuracy with 500 target samples, which is better than the 78% achieved from training from scratch. Advanced adversarial domain adaptation helps to reduce distribution differences. Multilingual joint training carried out with a common encoder and separate prediction heads for each language allows for zero-shot recognition through the use of phonological decomposition and semantic embeddings.

8.3. Linguistically-Informed Architectures

Using language-based models instead of purely data-driven ones is a justification for sustained precision and performance. Phonological decomposition represents the constituent parts of signs (about 50 handshapes × 30 locations × 20 movements × 8 orientations) instead of treating them as distinct groups, and this even allows generalization to unseen signs.

The rules applicable to ISL (for example, topic-comment structure) can be treated as either hard or trainable constraints that enhance the quality of continuous translation. Morphological modeling treats verb direction inflections and aspectual changes as integral parts of the main model.

8.4. ISLR-Bench Standardization Framework

Lack of standardization leads to a situation where the findings of studies lack direct comparability. We propose the ISLR-Bench as a conceptual framework for improving standardization in future ISL research. The first track, the Isolated Sign Recognition Track, measures accuracy on known signers and the critical unknown-signer generalization gap using the INCLUDE dataset. The Continuous Sign Recognition Track conducts Word Error Rate calculation in both controlled and in-the-wild environments using the CISLR dataset. The Regional Language Robustness Track suggests evaluating two regional variants in terms of cross-lingual transfer efficiency. The Real-Time Deployment Assessment Track reports latency and memory usage across various hardware, including desktop GPUs and mobile CPUs.

8.5. Five-Year Implementation Roadmap (2026 to 2030)

We propose a phased roadmap that could guide the translation of research into real-world impact. The first phase (2025–2026), Foundational Data and Benchmarks, creates a national ISL dataset collected through ethical methods and provides resources for five primary regional languages that DHH teams continuously monitor while generating public benchmarks and maintaining ongoing leaderboards. The second phase (2026–2027), Algorithmic Breakthroughs, aims at human-level recognition in controlled settings, demonstrating more than 99% isolated sign accuracy and less than 5% word error rate for continuous ISL performance. This phase creates advanced open-source models that come with specific constraints on processing speed and model size. The third phase (2027–2028), Real-World Robustness, operates extensive pilot programs across 15 states that test the systems with more than 10,000 DHH users while using unidentified signers and extreme weather to assess system performance. The fourth phase (2028–2029), Pervasive Deployment, enables complete real-time ISL recognition across consumer devices, achieving sub-50 ms latency on mid-range smartphones. The ISL Mobile SDK enables the creation of 500+ applications that will serve more than 1 million users. The fifth phase (2029–2030), Ecosystem Maturity and Equity, delivers complete public access to government digital services operating as official digital platforms, while evaluating success through three specific metrics: 15–25% employment growth for DHH workers, 70% literacy achievement, and the development of a sustainable community-based technology sector.

9. Synthesis and Discussion

This systematic review documents the complete progress of Indian Sign Language recognition technology that developed from 2020 to 2025, showing both major advances and ongoing challenges that prevent equitable implementation.

9.1. Persistent Barriers to Equitable Implementation

The current development of technology shows rapid progress, yet existing barriers still prevent people from accessing its full benefits. The continuous recognition process demonstrates an 8–15% performance decrease in actual situations because of co-articulation, temporal segmentation, and the addition of non-manual signals that include facial expressions and body posture. Multiple studies demonstrate that these signals play vital roles in understanding sign language. The regional language development process suffers from data availability problems because indigenous languages exist in datasets only 10–100 times smaller than ISL and at least 10 times smaller than ASL resources, which results from cost, coordination, and ethical restrictions. The issue of generalization is still present: for unknown signers there is a drop in accuracy of 5–18%, and a change in environment produces degradation of 8–15%, which indicates the necessity for diverse datasets and sophisticated training methods.

9.2. Stakeholder Call to Action

We recommend that stakeholders work together to support equitable ISL technology development. Researchers should prioritize real-world challenges, particularly continuous recognition, and ensure that evaluation includes unknown signers. They should also release code and models within six months of publication and promote the meaningful inclusion of DHH contributors, ensuring the presence of at least 50% DHH co-authorship.

Government bodies could consider setting up an allocation of INR 100 crore for the development of datasets, infrastructure, and innovation in 2026–2030; requiring all digital platforms to support ISL by 2028 with penalties for non-compliance; and creating a National ISL Data Commons.

9.3. Vision for 2030

We envisage that by 2030, ISL technologies could significantly enhance innovation and digital accessibility. Indian Sign Language and more than ten regional languages will be translated in real time with very good accuracy everywhere and on any device for both public and private usage. Technological barriers will diminish, which in turn will lead to growth in DHH community participation across different sectors such as employment (+15–25%), education (+10–15%), and healthcare access (+25–30%). The country's linguistic diversity will also be documented and acknowledged.

9.4. Technology As Bridge, Not Replacement

Technology should be regarded as an access enabler, not as a human replacement. It offers users empowerment through wide, constant, low-cost daily communication access. Still, it is not able to comprehend cultural intricacy, show emotional compassion, or make moral decisions like humans, particularly in delicate fields such as law or mental health. The future is hybrid, where technology takes over routine, high-volume interactions, allowing human professionals to deal with the delicate and complex communications, and where both roles are equally valued and rewarded socially.

9.5. Methodological Limitations

This review has several methodological limitations. Although a structured PRISMA approach was followed, the adaptation of QUADAS-2 into a quantitative scoring system is not formally validated and may introduce subjectivity. The study achieved broad database coverage yet may not have found all relevant studies because of database indexing limitations. The different research methods used in the included studies, together with the various assessment techniques employed to evaluate results, create barriers that make it difficult to directly compare the performance evaluation results across studies.

10. Conclusion

This systematic review has detailed the technical maturity and application potential of Indian Sign Language recognition from 2020 to 2025. The field shows progress through architectural evolution that started with CNNs (79–92%), moved to pose estimation (89–94%, 5–7× faster), and finally reached transformers (91–98% controlled, 81–90% real-world). The research has achieved substantial progress through its investigation of nine regional languages, which have been implemented in both healthcare and e-governance systems. The primary problems persist because an 8–15% performance loss occurs during continuous recognition, unknown-signer accuracy drops between 5–18%, and regional variant data remains 10–100 times scarcer. The desired results will depend on collaborative work between researchers, policymakers, industry experts, and members of the DHH community. The technological opportunity for truly equitable accessibility is not only a matter of debate but also one of social responsibility. To realize this vision, the collective will must be strong enough to build socially the strongest technical foundations, benefiting millions of DHH people in India.

Author Contributions

Gurusiddappa Hugar: Conceptualization, Methodology, Formal Analysis, Investigation, Writing (Original Draft), Writing (Review and Editing), Visualization. **Ramesh M. Kagalkar**: Validation, Data Curation, Formal Analysis, Investigation, Writing (Review and Editing).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that support the findings of this study are available within the article and its supplementary material.

Declaration of Generative AI and AI-Assisted Technologies In the Writing Process

During the preparation of this work the authors did not use any generative AI or AI-assisted technologies.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethics Statement

This systematic review synthesizes previously published research and does not involve human participants or animal subjects. All included studies obtained appropriate ethical approvals from their respective institutional review boards.

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